Generative AI in research and teaching

A practical and critical workshop, in two parts

Part 1 · Conceptualising the use of AI in research (12:00)

Part 2 · Practical AI for research and teaching (13:15)

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Before we begin

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Two parts, with a break between

When	Session	
12:00–12:30	**Part 1 · Conceptualising** the use of AI in research	the ideas
12:30–13:15	*Networking lunch*	*a break*
13:15–13:45	**Part 2 · Practical** AI for research and teaching	applied session

Part 1 sets up a way of thinking, the lunch is a real break and Part 2 puts the thinking into practice.

Part 1 · Conceptualising the use of AI in research

12:00–12:30 · why we are here, and the stance we will take

Our stance is discernment, not enthusiasm

- Whatever your level of confidence, whether you have never tried one or use them daily, you are in the right room.
- The aim is a deeper understanding of the technology, with no technical instruction on any one tool.
- This is a session about judgement. We use these tools and we interrogate them, in research and in teaching.
- Success means a clear account of what helped, what failed and where a human must stay in charge.

> “ *You bring the expertise. The tool is a fast and confident assistant whose error rate depends on the task, and whose fluency never tells you which case you are in. The aim is to leave knowing when not to reach for it.* ”

A brief word with your neighbour

In one sentence each, recall the last time an AI tool either helped you or let you down, and which it was.

Thirty seconds each. There is no need to perform. 'I have not really used one' is a perfectly good answer.

Most of us already have a story, and it usually turns on a moment of friction.

The latest in a long line of tools

Humans have always extended themselves with tools, and a labour economy that rewards productivity keeps moving us on to the next one.

- mental arithmetic → the calculator
- pen and paper → the computer
- the ledger → the spreadsheet
- the typewriter → the word processor
- the library catalogue → the search engine
- deterministic software → generative AI

> “ *This is the first entry on the list whose output can pass for judgement, so it earns more scrutiny than the last one did.* ”

Getting started, with no expertise needed

A generative AI tool works much like a chat assistant. You type a request and it writes back. That is essentially the whole interface.

- Choose one free tool today, such as ChatGPT, Claude, Gemini or Copilot. Any of them will serve.
- A useful request states who you are, what you want and what a good answer would look like. You then push back on the reply.
- Starter prompts for each track are in your group pack, ready to adapt.

> “ *Mixed groups are a strength. Newcomers ask the questions that matter, and regular users show what is possible. No one is behind.* ”

Cognitive friction is a signal

When a tool resists your intent, by giving a glib answer, missing the point or smoothing over a hard distinction, that friction is information. It marks where your judgement is doing real work.

- In a survey of knowledge workers, those with more confidence in the AI reported less critical-thinking effort, while those with more confidence in their own expertise reported more (Lee et al., 2025). Self-reported, and correlational.
- Deliberately reintroducing friction, through prompts that question the output, can restore critical and metacognitive engagement (Drosos et al., 2025).

We treat friction as data to record, whether we are marking, writing or analysing.

Deliberate friction, the 'cognitive forcing functions' that make you pause and decide, measurably cuts over-reliance on AI where explanations alone do not (Buçinca et al., 2021; link in the close).

The tool spectrum, from convenience to control

Level	Examples (free tiers)	You trade…
Off-the-shelf chat	ChatGPT, Claude, Gemini, Copilot	convenient · least control · data leaves your hands
No-code / low-code	Claude Projects, Gemini Gems, saved reusable prompts	some setup · repeatable · clearer guardrails
IDE / API level	Colab, Gradio/Streamlit, VS Code + assistant, AI Studio API key	steepest · most control · most responsibility

Effort and setup cost rise as you move down the table, and so does the control you keep over what the tool does with your data.

No-code means setting things up by clicking and prompting, without writing code. Working at the IDE or API level means using a code editor or calling the model programmatically, with more control and more responsibility.

Data-security red lines

Red line. If you would not pin it to a public noticeboard, do not paste it into a free consumer AI tool.

Personal data needs a lawful basis and a cleared tool. Confidential material is not yours to release at all. So never paste:

- Personal data of identifiable people, such as students, applicants, participants or staff (UK GDPR).
- Special-category data, such as health, ethnicity, political or religious beliefs, trade union membership, sexuality, genetics or biometrics.
- Confidential or unpublished material, such as grant drafts, peer-review files or data under an NDA.
- In teaching, also marks, references, pastoral notes, SpLD records, exam scripts and admissions data.

The ethical landscape

These risks are distinct, and they pull in different directions.

- **Accuracy**, fluent and confident yet wrong
- **Bias**, inherited from training data
- **Transparency**, when must use be disclosed?
- **Attribution and IP**, whose words and whose licence?
- **Equity**, who benefits and who is left out?
- **Environment**, an energy cost that scales with use
- **Over-reliance**, a quiet de-skilling
- **Accountability**, which never transfers

In teaching, add assessment integrity, disclosure to students and equitable access. Frameworks (Russell Group, EC, UNESCO, ICO) in the close.

Stepping back, the bigger questions

Beyond any single task, three larger shifts are worth carrying into Part 2.

- Thinking and writing, decoupled. We have long thought by writing. If the tool drafts, where does the thinking go, and what is lost if we let it?
- Fairness, as leveller or amplifier. AI might narrow gaps (a second language, thin support, an early career) or widen them. Which, and for whom?
- Disclosure. We ask people to disclose their use of AI. Should we also disclose what it offset for us, such as writing in a second language or having little support, or would that expose the very inequities it eased?

> *There are no settled answers here. These are policy questions in the making. Notice them in your own task.*

The human-in-the-loop principle

The AI proposes and the human disposes. Accountability does not transfer to a tool.

Before you start a task, decide the following in advance.

1. Where must a human verify a claim, a number or a citation?
2. Who signs off, and against what standard?
3. What will you never delegate, such as the final call on correctness, ethics or authorship?
4. How is oversight arranged? Interwoven means a human checks at every step. Staged means a human checks at a few points between phases.

> “ *A good workflow names its checkpoints before it trusts the output, not after something has broken.* ”

What Part 2 asks of you

Take a real task, use a tool on it, then examine the result closely. Part 1 provided the way of thinking. Part 2 puts it into practice. You will look at your work through five dimensions, and score them in your pack afterwards if you wish.

- Project Definition, your group's one real problem.
- Technology Stack, which tool on the spectrum, and why (Part 1).
- Data Security & Ethics, the red lines you hold (Part 1).
- Financial & Scalability, whether the free tier holds.
- Human-in-the-Loop, where you steer, interwoven or staged (Part 1).

Plus the friction you catch (your museum of caught errors), a short societal reflection (fairness, disclosure, thinking versus writing and whether your field is over- or under-using AI here) and one insight to share. The app captures the essentials, and the rubric and the reflection are in your pack.

Form your groups now

Groups of five. Move to the idea you would most like to try. The letter beside it is your track.

- Stress-test a study design (A)
- Critique an assessment or rubric (A)
- Turn messy notes into clear actions (B)
- Triage an email or admin backlog (B)
- Plan a module, project or paper (B)
- A one-page explainer of a finding (C)
- A quick visualisation or teaching aid (C)
- A lay summary or press angle (D)
- Explain a concept for students (D)
- A social thread from a paper (D)

Note who is in your group and which track you chose. You regroup straight after lunch, without re-forming.

A · Methodological Blind-Spot Detector, turn the tool on your own design and verify its critiques. B · Executive-Function Layer, build a reusable aid for planning or triage. C · Rapid Prototyping, make a small artefact, then check it. D · Public Engagement, translate a finding, then audit it for fidelity. Fuller briefs are in your group pack, with one-line summaries in the app. Your group takes one problem into Part 2, either this seed or a real one a member brings.

Networking lunch

12:30–13:15

Part 2 · Practical AI for research and teaching

13:15–13:45 · applied session · groups of five

Open the workshop app

`genai-rt.web.app`

Scan the code, or type the address into any browser.

Your facilitator will read out the session passcode you need to start a group.

No device, or trouble connecting? HackMD (hackmd.io) is the fallback, and your facilitator will point you there.

How Part 2 runs (30 minutes)

When	Phase	Mins
13:15	**Settle in.** Re-find your group, agree your one problem (60 s), open the app (HackMD as fallback), red lines on	3
13:18	**Apply it.** Run the tool on your problem and capture one caught error and one insight (the automation–steering map if time allows)	15
13:33	**Lightning round.** 45 seconds per group, on the limitation, the safeguard or how your field over- or under-uses AI	7
13:40	**Synthesis.** Drawing the threads together, with takeaways for research and teaching	5

The five angles

Your pack rates your work on five dimensions, from 1 (Nascent) through 3 (Developing) to 5 (Robust). Share the five angles across the group and pass the keyboard freely. The scoring is for your pack and the discussion. The app holds the essentials.

1. Project Definition (Convenor). Your group's one real problem, kept in focus.
2. Technology Stack (Driver). The right tool on the spectrum, and why.
3. Data Security & Ethics (Steward). The red lines you hold.
4. Financial & Scalability (Reporter). Whether the free tier holds. This role also owns the note and the spoken insight.
5. Human-in-the-Loop (Sceptic). Checkpoints and accountability, plus the automation–steering map if time allows.

The five dimensions frame your reflection and the fuller fallback note. During the session, focus on the core three. An experienced group can ignore the labels.

Working on a real problem

Use your group's one problem: the seed, a member's real one or a worked example from your pack. Complete these three first. If that is all you manage, you have done the task.

1. One artefact. Run your tool and produce something real, such as a critique, a template or a translation.
2. One or two caught errors. Moments where it was fluent but wrong. Keep them.
3. One insight. A limitation, a safeguard or one honest observation about how your field is over- or under-using AI. This is your line for the 45-second round.

> “ *If about five minutes remain, go further. Add the automation–steering map (which steps the tool ran, and where you steered) and decide whether oversight was interwoven or staged, noting why and what it cost. If you fall behind, drop the map before the insight.* ”

The map and the oversight model are a bonus if time allows. The rubric scores and the societal reflection are for your pack and the discussion.

Capturing your work in the app

There is no GitHub to touch and no report to write. This is fifteen minutes of capture, kept light.

1. The Reporter opens the app (genai-rt.web.app), names the group and enters the session passcode. There is no account and nothing to install. One device creates the group, then reads out the group name and the six-character code, and the others type both.
2. Fill in the essentials: the problem, the artefact (and which tool made it), one caught error and your insight, adding the map and the oversight model if time allows. The app saves as you go.
3. Submit for review. The facilitator sees it live and approves it. That is all.

No device, or the app misbehaving? HackMD (hackmd.io) is the fallback, with the same headings. A countdown on screen shows the time left, and the facilitator archives the approved work afterwards.

The museum of caught errors

Keep a short, honest log of the moments the tool went wrong, and what each one taught you.

- A confident but fake citation · a flattened nuance · a plausible but invalid method · a biased rewrite · a misgraded answer.
- For each, note what happened, how you caught it and what it signals about where humans must stay in charge.

> “ *These entries are the most valuable thing your group produces.* ”

Fifteen minutes, starting now (13:18)

Tracks are in your pack and in the app · the rubric is in your pack · help is circulating

Hold the red lines, record the friction, keep a human in the loop.

⏱ Pivot, around 13:26

Thirty seconds, keyboards down, as a group.

- What has the tool genuinely made easier, and what does that cost?
- Where did it resist you, and what was that friction telling you?
- Have we held the data red lines?
- Switch now from building to interrogating. Settle your map and your insight.

Lightning round · 13:33

45 seconds per group · one insight · no slides · the Reporter delivers, the facilitator keeps time

What was the most significant limitation, the most important human-in-the-loop safeguard, or one honest thing this exposed about how your field over- or under-uses AI?

Close · 13:40

Drawing the threads from the room, and what we take back to research and teaching

What we saw across the room

Four threads worth drawing out from the lightning round:

- **Context decided.** A tool that helped on one task and was a hazard on the next.
- **Friction as a signal.** Where the friction a group recorded marked where judgement lives.
- **A checkpoint named in advance.** The strongest safeguard a group built in.
- **Where a free tier drew a hard line.** On data, scale or accountability.

The through-line

> “ *Use the tool where it earns its place. Keep the human where judgement is owed.* ”

- Treat fluency as a reason to check.
- Build the checkpoint before you trust the output.
- Record the friction. It maps where you matter.
- Disclose use, respect data and keep accountability with a named person.
- Ask the bigger questions of who is helped, who is left out and what we owe to disclose.

Back at your desk

Choose one to try this week, in research or in teaching.

1. Write a two-line note on your AI use for one task, recording what you used, what you checked and what you did not delegate.
2. Add one human checkpoint to a workflow you already assist with AI.
3. Before pasting anything sensitive, work through the one-page data decision aid from your pack.
4. Keep your own museum of caught errors for a fortnight.

A short follow-up, with a responsible-use commitment, will reach you by email.

Further reading

- **Russell Group (2023).** Principles on generative AI in education. russellgroup.ac.uk
- **EC / ERA Forum (2026, third version).** Living guidelines on the responsible use of generative AI in research.
- **UNESCO (2023).** Guidance for generative AI in education and research (Miao & Holmes).
- **ICO.** Guidance on AI and data protection (UK GDPR). ico.org.uk
- **Lee et al. (2025, CHI).** The impact of generative AI on critical thinking, a survey of knowledge workers. doi.org/10.1145/3706598.3713778
- **Drosos et al. (2025).** 'It makes you think': provocations help restore thinking. arXiv:2501.17247
- **Buçinca et al. (2021, CSCW).** Cognitive forcing functions reduce over-reliance on AI. doi.org/10.1145/3449287
- **Bender et al. (2021, FAccT).** On the dangers of stochastic parrots, fluent remixing without understanding. doi.org/10.1145/3442188.3445922
- **Karpathy (2025).** *Software is changing (again)*, the human on the 'autonomy slider' [talk].
- **Mollick (2023).** *Centaurs and cyborgs*, the human stays the architect [essay].

Full citations, with links, are in the repository (github.com/pablobernabeu/genai-research-teaching).

Thank you

Your notes become an open, reproducible archive at github.com/pablobernabeu/genai-research-teaching

Questions or follow-ups? Open a thread in the repository's Discussions tab (github.com/pablobernabeu/genai-research-teaching/discussions), so that answers help everyone.

The tool is fast and confident. You are the one who is accountable, and who decides when its fluency has earned your trust. Bring both to the work.

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